



VACUTAP[®] VVS[®]

COMPACT ON-LOAD TAP-CHANGER
FOR GRID APPLICATIONS.

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VACUTAP® VVS® – TAILORED FOR GRID APPLICATIONS.

The VACUTAP® VVS® vacuum on-load tap-changer was developed and optimized for operation in transformers up to approximately 60-70 MVA for 110 kV* power grids. Its compact design combines a diverter switch and tap selector in one functional unit.

The VACUTAP® VVS® is based on the well-proven VACUTAP® VV® and comprises a diverter switch and selector within the oil compartment. This compact design enables a spatially optimized configuration of the transformer and transformer cell.

Maintenance-free for nearly all grid applications

The vacuum-switching technology already proven in the VACUTAP® VV® ensures maximum reliability. The maintenance interval of 300,000 operations means maintenance-free performance for nearly all grid applications and therefore maximum availability of your transformers. This is made possible through features such as vacuum interrupters, which are

developed in accordance with our strict requirements and have proven themselves in years of operation. No other manufacturer of tap changers has such a large number of vacuum on-load tap-changers in operation as Maschinenfabrik Reinhausen.

Benefits of vacuum technology

- Substantially reduced frequency of maintenance compared to conventional oil circuit engineering
- No arcing in the tap-changer oil
- No oil filter unit needed

*voltage levels may vary by country



VERY COMPACT. MOST RELIABLE.

The on-load tap-changers of the VACUTAP® series combine decades of experience in vacuum-switching technology with the consistent further development of this technology. This results in on-load tap-changers with maximum reliability and availability.

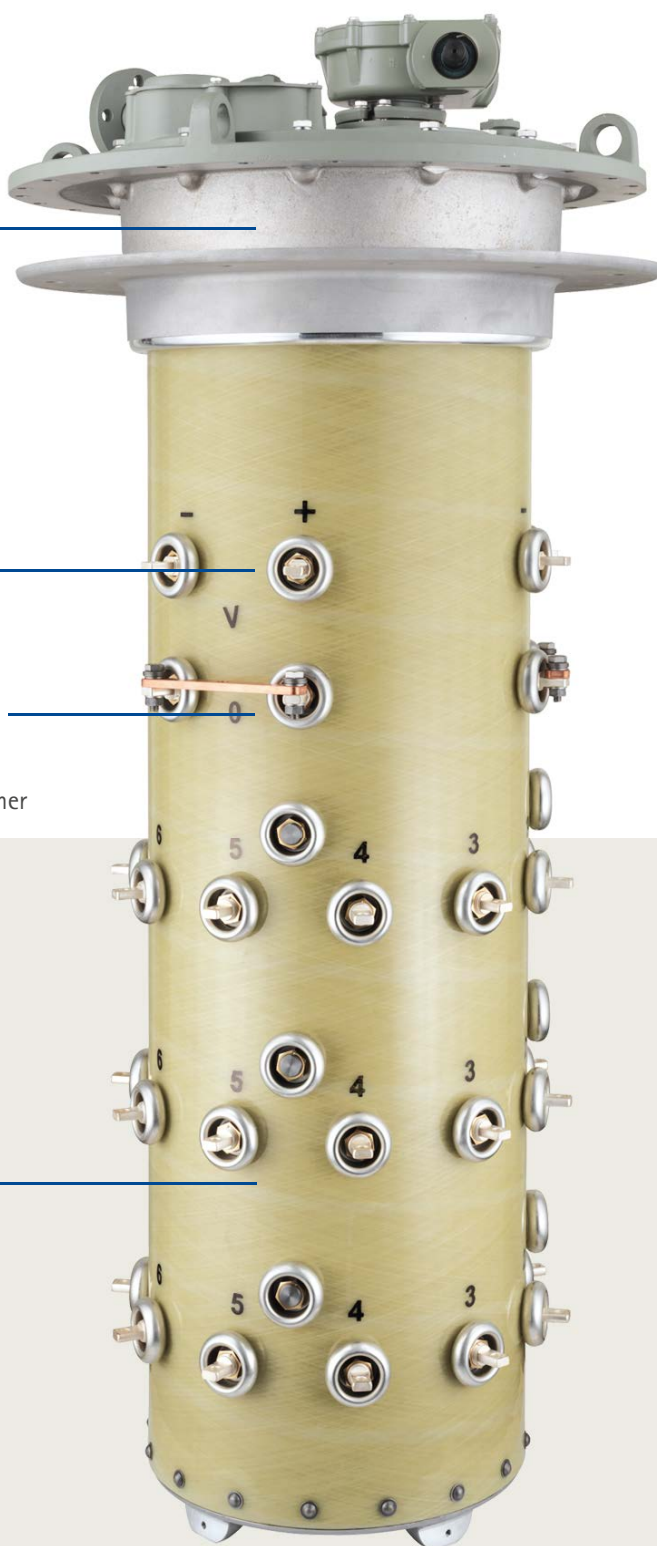
Flange for bell-type tank installation
standard to simplify assembly

Enhanced change-over selector capacity,
usually no potential connection measure needed

Internal integrated change-over selector

- Optimized dimensions
- No influence on the DGA values of the transformer

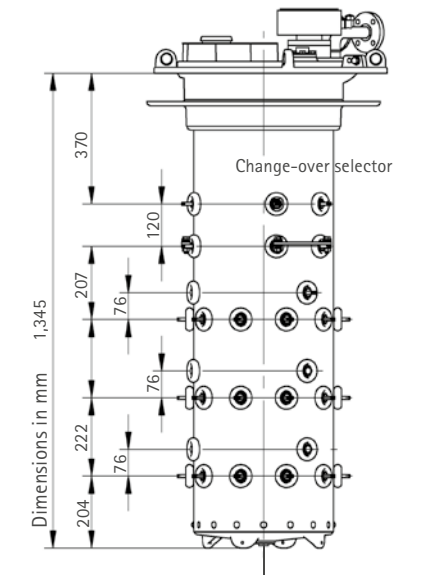
Space saving in the transformer tank due
to very compact dimensions designed for
standard grid applications



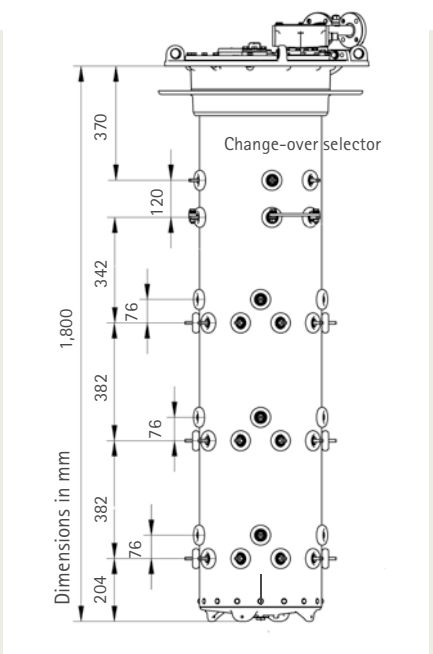
VACUTAP® VVS®

TECHNICAL DATA.

Installation length VVS® III 400 Y

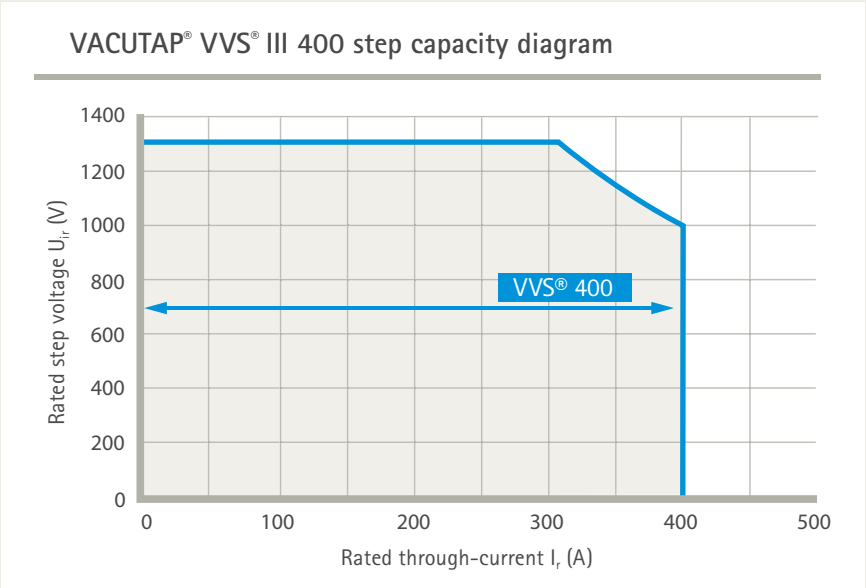


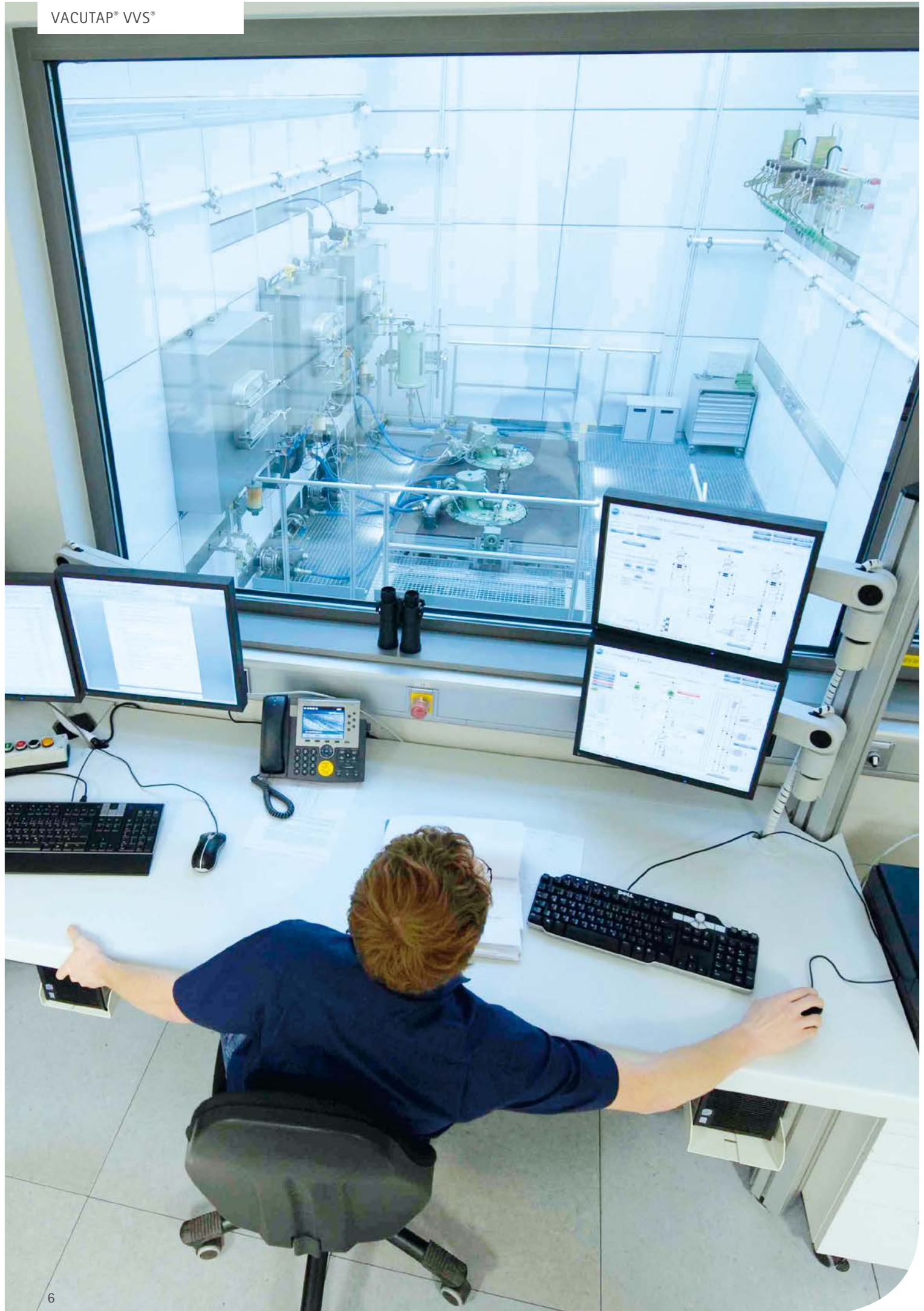
Installation length VVS® III 400 D



On-load tap-changer	VVS® III 400 Y	VVS® III 400 D
Number of phases	3	
Max. rated through-current I_r (A)	400	
Rated short-time current (kA)	4	
Rated duration of short-circuits (s)	3	
Rated peak withstand current (kA)	10	
Max. rated step voltage U_{ir} (V)	1,300	
Step capacity P_{stN} (kVA)	400	
Rated frequency (Hz)	50 to 60	
Operating positions	with change-over selector: max. 19	
Motor-drive unit	TAPMOTION® ED, ETOS® ED	

Rated insulation level	VVS® III 400
Highest voltage for equipment U_m (kV)	40 or 76
Rated lightning impulse withstand voltage (kV, 1.2/50 μ s)	200 or 350
Rated short-duration power frequency withstand voltage (kV, 50 Hz, 1 min.)	70 or 140





HIGH QUALITY. LOW LIFE-CYCLE COSTS.

For reliable, economical operation.



Maintenance-free and long-lasting

- Maintenance interval of 300,000 tap-changes without time-based components
- Maintenance free for nearly all grid applications



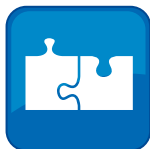
Maximum operational reliability

- In our globally unique testing department, our on-load tap-changers are tested over all technical limits
- No other manufacturer of tap-changers has such a large number of vacuum on-load tap-changers in operation worldwide



Low life-cycle costs

- With this product we offer the best price-performance ratio
- Maintenance-free in most cases during the transformer life time



Compact design

- The innovative design is especially well-suited for applications with little available space
- The integrated change-over selector enables a more effective transformer design that uses less material



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THE POWER BEHIND POWER.

